



Self Development Tracker

Functional Requirements Specification (FRS)

Version 1.0

SDT V1 - Local Application

Prepared by: Kyaw Zay Han

Role: Business Analyst / Project Owner

27 August 2026

Document Control

Field	Details
Document Title	SDT - Functional Requirements Specification
Project	Self Development Tracker (SDT)
Version	1.0
Application Version	SDT V1 - Local Application
Date	27 August 2026
Status	Final
Parent Document	SDT Business Requirements Document (BRD) v1.0
Purpose	Translate approved business requirements into detailed, testable system behaviour for SDT V1.

1. Purpose

This Functional Requirements Specification (FRS) defines what SDT V1 must do from a user and system-behaviour perspective. It translates the high-level business requirements established in the Business Requirements Document into detailed functional requirements that can be implemented, verified and traced to test cases.

The specification focuses on observable behaviour and required outcomes. Low-level implementation choices such as individual API endpoints, SQL statements and source-code structure are intentionally excluded unless they are necessary to define required system behaviour.

2. Functional Scope

The functional scope covers the complete SDT V1 challenge lifecycle for a single local user:

- Challenge setup and configuration.
- Rule/task definition before challenge start.
- Active challenge dashboard and calendar.
- Daily checklist recording and editing.
- Today shortcut for faster daily entry.
- Overall progress calculation.
- Per-rule strike tracking.
- In-progress challenge reporting.
- Manual challenge ending.
- Natural challenge completion and final reporting.
- Local data persistence and user-facing error handling.

Authentication, multi-user support, public cloud hosting, social features, automated reminders and third-party fitness integrations remain outside the SDT V1 functional baseline.

3. Actors and System Context

Actor	Description	Primary Interactions
End User	The individual using SDT to manage a self-development challenge.	Configure challenge; define rules; start/end challenge; record daily adherence; review dashboard and reports.
SDT Application	The local web application that applies challenge rules and manages the user workflow.	Validate input; persist records; calculate progress/strikes; render calendar; generate reports.
Local Database	Persistent relational data store used by SDT V1.	Store challenge, task/rule and daily status data.

4. Requirement Conventions

Each functional requirement is assigned a unique FR identifier. The Parent BR column identifies the originating business requirement from the BRD. Priority indicates relative importance to the V1 solution.

Notation	Meaning
FR-XX	Functional Requirement identifier.
BR-XX	Parent Business Requirement from the BRD.
BRULE-XX	Business Rule governing system behaviour.
High	Essential to the intended V1 operation.
Medium	Important but not fundamental to the core tracking lifecycle.
Low	Desirable enhancement if applicable.

5. Challenge Setup and Configuration

ID	Parent BR	Priority	Functional Requirement	Acceptance / Verification
FR-01	BR-01	High	When no active challenge exists, the system shall display the challenge setup screen rather than the active dashboard.	Setup screen is visible and the active dashboard is not shown.
FR-02	BR-01	High	The system shall allow the user to enter a challenge name.	A non-empty challenge name can be entered and retained for challenge creation.
FR-03	BR-01	High	The system shall allow the user to select a challenge start date.	A valid start date can be selected before challenge creation.
FR-04	BR-01	High	The system shall allow the user to specify the challenge duration in days.	A positive duration within the supported range can be entered.
FR-05	BR-07	High	The system shall allow the user to specify the number of strikes allowed per rule.	A non-negative strike allowance can be configured.
FR-06	BR-02	High	The system shall allow the user to add one or more rules/tasks before the challenge starts.	New rule/task input rows can be created through the setup interface.
FR-07	BR-02	High	The system shall allow the user to remove rules/tasks before the challenge starts.	A selected setup rule can be removed without editing the database.
FR-08	BR-02	Medium	The system shall allow the user to reorder rules/tasks before the challenge starts.	The displayed rule order can be moved and retained when the challenge is created.
FR-09	BR-01	High	The system shall prevent challenge creation when required setup information is missing or invalid.	A validation message is shown and no invalid challenge is created.
FR-10	BR-01	High	The system shall require at least one non-empty rule/task before a challenge can start.	Challenge creation is rejected when the final rule list is empty.
FR-11	BR-01	High	When the user starts a valid challenge, the system shall persist the challenge configuration and its rules/tasks.	The new challenge becomes active and remains available after application restart.
FR-12	BR-03	High	After a challenge starts, the active dashboard shall not provide controls to add, remove or reorder the challenge's rule set.	The rule baseline remains stable during the active challenge.

6. Active Challenge Dashboard

ID	Parent BR	Priority	Functional Requirement	Acceptance / Verification
FR-13	BR-05	High	When an active challenge exists, the system shall display the active challenge dashboard.	The dashboard is shown with the current challenge rather than the setup screen.
FR-14	BR-05	High	The dashboard shall display the active challenge name, start date, end date, duration, number of rules and strike allowance.	Displayed metadata matches the stored challenge.
FR-15	BR-06	High	The dashboard shall display an overall challenge progress indicator.	A percentage/progress bar and completed-day count are visible.
FR-16	BR-07	High	The dashboard shall display strike information for each configured rule.	Each rule shows strikes used and/or remaining.
FR-17	BR-15	High	The dashboard shall provide a Today shortcut.	A Today control is visible during an active challenge.
FR-18	BR-10	High	The dashboard shall provide access to the active progress report.	The user can open the progress report from the dashboard.
FR-19	BR-12	Medium	The dashboard shall provide an End Challenge action.	The user can initiate manual challenge termination.

7. Calendar and Challenge Date Control

ID	Parent BR	Priority	Functional Requirement	Acceptance / Verification
FR-20	BR-05	High	The system shall render a calendar view for the active challenge.	Calendar cells are displayed for the selected month.
FR-21	BR-05	High	For dates within the challenge period, the calendar shall display the corresponding challenge day number in the form Day X/Y.	The day number is calculated from the start date and total duration.
FR-22	BR-05	High	Dates before the challenge start date and after the challenge end date shall be visually distinguished as outside the challenge.	Outside dates are greyed or otherwise disabled.
FR-23	BR-05	High	Dates outside the challenge period shall not open a daily checklist.	Clicking an outside date does not allow progress entry.
FR-24	BR-05	Medium	The user shall be able to navigate to the previous and next calendar month.	Month navigation updates the rendered calendar.
FR-25	BR-05	Medium	The system shall visually identify today when today falls within the challenge period.	Today's valid challenge cell is distinguishable from other cells.
FR-26	BR-06	High	For valid challenge dates, the calendar shall display recorded completion information for that day.	The calendar cell reflects stored daily completion.

8. Daily Checklist Recording

ID	Parent BR	Priority	Functional Requirement	Acceptance / Verification
FR-27	BR-04	High	Selecting a valid challenge date shall open a Day Details dialog for that date.	The dialog identifies the selected date and challenge day number.
FR-28	BR-04	High	The Day Details dialog shall list every configured challenge rule/task as a checkbox.	The checklist corresponds to the stored rule set.
FR-29	BR-04	High	The system shall load previously saved completion states when a recorded day is reopened.	Saved checkbox states are restored accurately.
FR-30	BR-04	High	The user shall be able to tick or untick any listed rule/task before saving.	Checkbox state can be changed in the user interface.
FR-31	BR-04	High	The Day Details dialog shall explain that the user should tick/untick completed tasks or followed rules and then save.	Instructional text is visible in the dialog.
FR-32	BR-04	High	When Save is selected, the system shall persist the completion state of all rules/tasks for the selected date.	Reopening the date returns the saved states.

FR-33	BR-04	High	After a successful save, the system shall refresh the calendar, overall progress and strike information.	Dashboard values reflect the newly saved data.
FR-34	BR-15	Medium	After a successful save and dashboard refresh, the Day Details dialog shall close automatically.	The user returns to the dashboard without manually closing the dialog.
FR-35	BR-04	High	If saving fails, the system shall keep the dialog available and communicate the failure rather than treating the record as saved.	No false success state is presented.

9. Today Shortcut

ID	Parent BR	Priority	Functional Requirement	Acceptance / Verification
FR-36	BR-15	High	When Today is selected and the current date is within the active challenge period, the system shall navigate to the current month and open today's Day Details dialog.	Today's checklist opens directly.
FR-37	BR-15	High	When Today is selected and the current date is outside the active challenge period, the system shall prevent progress entry and inform the user.	No invalid day record can be created through the shortcut.

10. Progress and Completion Logic

ID	Parent BR	Priority	Functional Requirement	Acceptance / Verification
FR-38	BR-06	High	The system shall calculate daily completion from the number of completed rules/tasks relative to the total configured rules/tasks.	Daily completion reflects stored task states.
FR-39	BR-06	High	The system shall identify a perfect recorded day when all configured rules/tasks are marked complete.	Perfect-day status is derived only from a recorded day with 100% completion.
FR-40	BR-09	High	The system shall distinguish an unrecorded challenge day from a recorded day containing incomplete rules/tasks.	Missing records are not automatically treated as failed records.
FR-41	BR-09	High	The system shall not consume strikes solely because a challenge day has not been recorded.	Strike counts remain unchanged for unrecorded days.
FR-42	BR-06	High	The system shall calculate overall progress using the application's defined completed-day logic and total challenge duration.	Dashboard and report values remain consistent with stored daily records.

11. Strike Tracking

ID	Parent BR	Priority	Functional Requirement	Acceptance / Verification
FR-43	BR-07	High	For each recorded day, the system shall treat each unchecked rule/task as a missed occurrence for that rule.	Missed counts derive from recorded checklist states.
FR-44	BR-07	High	The system shall calculate strikes independently for each rule/task.	One rule's misses do not directly change another rule's strike count.
FR-45	BR-07	High	The system shall display strikes used and strikes remaining for each rule/task.	Displayed values match the configured allowance and recorded misses.
FR-46	BR-08	High	When strikes used equal or exceed the configured allowance, the system shall continue the challenge without automatic restart or termination.	Challenge status remains active unless otherwise completed or ended.
FR-47	BR-08	High	The system shall permit the recorded missed count	Additional misses may be

			to exceed the configured strike allowance.	recorded even when strikes remaining is zero.
FR-48	BR-08	Medium	The dashboard shall communicate that strike tracking is informational and does not trigger penalties or resets.	The interface does not imply automatic punishment.

12. Progress Report

ID	Parent BR	Priority	Functional Requirement	Acceptance / Verification
FR-49	BR-10	High	The user shall be able to open a progress report for the current active challenge.	A report view is generated for the active challenge.
FR-50	BR-10	High	The progress report shall identify the challenge and its configured period.	Challenge name and relevant dates are displayed.
FR-51	BR-10	High	The progress report shall show challenge position, recorded days, unrecorded days and perfect days where applicable.	Summary metrics are derived from stored data.
FR-52	BR-10	High	The progress report shall show average recorded completion and completed/missed task counts.	Metrics are consistent with daily records.
FR-53	BR-10	High	The progress report shall show rule-level performance and strike usage.	Per-rule results are visible.
FR-54	BR-10	Medium	The progress report shall show daily performance information sufficient to review performance over time.	Daily results are represented in the report.
FR-55	BR-10	Medium	The progress report shall support printing or saving through the browser's print capability.	The report has a printable layout.

13. Manual Challenge End

ID	Parent BR	Priority	Functional Requirement	Acceptance / Verification
FR-56	BR-12	High	Selecting End Challenge shall display a confirmation dialog before the challenge is ended.	The challenge cannot be ended by the first click alone.
FR-57	BR-12	High	The confirmation dialog shall inform the user that existing challenge and progress data will be preserved.	The effect of the action is communicated before confirmation.
FR-58	BR-12	High	If the user cancels the confirmation, the active challenge shall remain unchanged.	No termination is recorded.
FR-59	BR-12	High	If the user confirms, the system shall mark the current challenge as completed/ended rather than deleting it.	Historical records remain available.
FR-60	BR-13	High	Manual termination shall preserve challenge configuration, rules/tasks and recorded daily performance.	Stored history remains available for reporting.
FR-61	BR-11	High	After successful manual termination, the system shall open the final report for the ended challenge.	The user is taken to the appropriate final report.

14. Natural Completion and Final Report

ID	Parent BR	Priority	Functional Requirement	Acceptance / Verification
FR-62	BR-11	High	The system shall recognise when a challenge has reached its configured natural completion condition.	The challenge can be identified as naturally completed.
FR-63	BR-11	High	A naturally completed challenge shall be eligible for a final report.	Final reporting is available without manual termination.
FR-64	BR-11	High	The final report shall use the completed challenge's stored configuration and daily performance data.	Report values are based on the relevant completed challenge.

FR-65	BR-11	High	The final report shall remain meaningful for both naturally completed and manually ended challenges.	Completion context does not prevent reporting.
FR-66	BR-11	Medium	The final report shall support browser printing or saving as PDF.	A print-friendly output is available.
FR-67	BR-13	Medium	After a challenge has ended, the application shall permit the user to return to challenge setup and create a new challenge while preserving prior stored history.	A new challenge can begin without deleting the previous completed challenge.

15. Data Persistence and Error Handling

ID	Parent BR	Priority	Functional Requirement	Acceptance / Verification
FR-68	BR-13	High	Challenge, rule/task and recorded daily status data shall persist after the Node.js application is stopped and restarted, provided the database remains intact.	Previously stored data reloads after restart.
FR-69	BR-14	High	SDT V1 shall store its operational data in the configured local database rather than relying on browser-only temporary storage for core challenge records.	Core records survive browser closure.
FR-70	BR-01	High	The system shall communicate request or data-access failures to the user or surface a clear application error rather than silently failing.	Failures do not appear as successful operations.
FR-71	BR-14	Medium	The application shall read local environment/database configuration without requiring credentials to be embedded in frontend source files.	Runtime configuration is separated from public client code.

16. Functional Business Rules Mapping

Rule	Summary	Related Functional Requirements
BRULE-01	Only one challenge may be active at a time.	FR-01, FR-11, FR-13, FR-67
BRULE-02	Rules/tasks are treated as locked after challenge start.	FR-12
BRULE-03 / BRULE-04	Only dates inside the configured challenge period are valid for tracking.	FR-21 to FR-26, FR-37
BRULE-05 / BRULE-06	Recorded days store rule states; 100% recorded completion is a perfect day.	FR-32, FR-38, FR-39
BRULE-07 to BRULE-10	Missed rules on recorded days drive per-rule strikes; unrecorded days do not; exceeding the allowance has no automatic penalty.	FR-40 to FR-48
BRULE-11	Today opens a checklist only within the challenge period.	FR-36, FR-37
BRULE-12 / BRULE-13	Manual termination requires confirmation and preserves history.	FR-56 to FR-60
BRULE-14	Final reporting is available after natural or manual completion.	FR-61 to FR-66

17. High-Level Requirements Traceability

Business Requirement	Functional Coverage
BR-01 - Challenge Configuration	FR-01 to FR-05, FR-09 to FR-11, FR-70
BR-02 - Rule Definition	FR-06 to FR-08, FR-10
BR-03 - Rule Stability	FR-12
BR-04 - Daily Recording	FR-27 to FR-35
BR-05 - Calendar Visibility	FR-13, FR-14, FR-20 to FR-26
BR-06 - Progress Monitoring	FR-15, FR-26, FR-38 to FR-42
BR-07 - Strike Monitoring	FR-05, FR-16, FR-43 to FR-45
BR-08 - Non-Punitive Strikes	FR-46 to FR-48

BR-09 - Unrecorded Day Treatment	FR-40, FR-41
BR-10 - Active Challenge Reporting	FR-18, FR-49 to FR-55
BR-11 - Completion Reporting	FR-61 to FR-66
BR-12 - Manual Completion	FR-19, FR-56 to FR-59
BR-13 - Historical Preservation	FR-60, FR-67, FR-68
BR-14 - Local Operation	FR-69, FR-71
BR-15 - Usability / Today Shortcut	FR-17, FR-34, FR-36, FR-37

18. Non-Functional Requirements Reference

The functional requirements operate alongside the non-functional baseline established in the BRD: usability, local performance, reliable persistence, data integrity, maintainability, secure environment configuration, desktop-browser compatibility and recoverability. These qualities should be validated during system and UAT testing even where they do not map to a single screen action.

19. Acceptance and Testing Approach

Each functional requirement is written to support verification. The subsequent Requirements Traceability Matrix (RTM) will map each business requirement to one or more functional requirements and then to corresponding test cases. UAT will focus on observable behaviour from the user's perspective, while technical testing may additionally validate persistence, data integrity and error handling.

A functional requirement is considered satisfied only when the expected behaviour is implemented and the corresponding test evidence passes. Defects should reference the affected FR identifier so that requirement impact remains visible.

20. Change Control

This FRS represents the SDT V1 functional baseline. New behaviour that changes agreed scope, business rules, user access, deployment architecture or data handling should be assessed through change control rather than silently added to the baseline.

When a change is approved, affected BR and FR identifiers should be updated or supplemented, and the Requirements Traceability Matrix and test cases should be revised accordingly.

21. Approval / Baseline

Baseline Status: Approved

Version: 1.0

Approved by: Kyaw Zay Han, Project Owner / Business Analyst

Date: 27 August 2026